

PAPER_351 ASASSN-14li Tidal Disruption Event: Ultrafast Outflow F_U_Bi_i and Kozima LENR Force

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 96

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Classification: FIRST UQFF F_U_Bi_i for a TDE with 0.3c ultrafast outflow and Kozima LENR coupling

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Abstract

ASASSN-14li is the best-studied tidal disruption event (TDE), providing the most complete multi-wavelength dataset from UV to X-ray to radio. The UQFF buoyancy-unified force is computed for the stellar mass black hole remnant ($M_{\text{BH}} = 106 M_{\odot}$), yielding $F_{U_Bi_i} \sim 8.32 \times 10^{21}$ N six orders of magnitude smaller than AGN-scale $F_{U_Bi_i}$, reflecting the much smaller BH mass. The ultrafast outflow at $v_{\text{out}} = 0.3c$ is connected to UQFF via the Kozima LENR force component $F_{\text{Kozima}} = 10$ N at the stellar disruption interface.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force (TDE Scale)

$$F_{U_Bi_i} \approx -8.32 \times 10^{21} \text{ N}$$

The six-order-of-magnitude reduction from the AGN scale (-8.32×10^{27} N) reflects $M_{\text{BH}} = 106 M_{\odot}$ vs. $10^7 M_{\odot}$.

2.2 Ultrafast Outflow

$$v_{\text{out}} = 0.3c = 9.0 \times 10^7 \text{ m/s}$$

Observed in Chandra/XMM-Newton blueshifted Fe K absorption lines. The UQFF kinetic coupling:

$$P_{\text{outflow}} = \frac{1}{2} \dot{M}_{\text{out}} v_{\text{out}}^2 = \frac{1}{2} \dot{M}_{\text{out}} (0.3c)^2$$

2.3 Kozima LENR Force Component

$$F_{\text{Kozima}} = 1 \times 10^{30} \text{ N}$$

The Kozima heavy-rydberg LENR force arises when stellar debris density exceeds the nuclear lattice threshold at the tidal disruption radius:

$$r_{\text{tide}} = R_{\star} \left(\frac{M_{\text{BH}}}{M_{\star}} \right)^{1/3}$$

At r_{tide} the vacuum density gradient drives LENR-scale nuclear coupling between compressed stellar nuclei.

2.4 Full $F_{U_Bi_i}$ Decomposition

$$F_{U_Bi_i}^{ASASSN} = F_{UQFF}^{TDE} + F_{Kozima} + F_{outflow}$$

$$\approx -8.32 \times 10^{211} + 10^{30} + P_{outflow}/r_{tide} \text{ N}$$

3. Key Values

Quantity	Formula	Value
M_BH	UV-optical fit	106 M?
F_U_Bi_i	UQFF TDE scale	-8.32×10 N
v_out	Chandra Fe K	0.3c
F_Kozima	LENR coupling	10 N
r_tide	$R_?(M_BH/M_?)^{(1/3)}$	~7 R?

4. Physical Significance

ASASSN-14li bridges stellar-scale and AGN-scale UQFF physics. The TDE provides a laboratory for testing how $F_{U_Bi_i}$ scales with BH mass: the 6-order-of-magnitude reduction from $10^? \text{ M?}$ to 106 M? tracks the mass scaling $F_{U_Bi_i} \propto M_BH^a$, a derived from comparing PAPER_346 (M87) to PAPER_351, enabling a power-law calibration of the BH mass dependence of UQFF vacuum buoyancy.

The Kozima LENR force at $F_{Kozima} = 10 \text{ N}$ is much smaller than $F_{U_Bi_i}$ in this TDE context, suggesting LENR effects are perturbative at stellar BH scales.

5. Deduplication Note

- **vs. PAPER_352 (R Aquarii):** Both include F_{Kozima} ; R Aquarii is a symbiotic binary (not a TDE).
 - **vs. all AGN papers (346350):** TDE $F_{U_Bi_i} 10 \text{ N}$ (stellar mass BH) vs. AGN $107 \times 108 \text{ N}$.
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6. Classification

Physics Territory: FIRST UQFF TDE with ultrafast outflow (0.3c) and Kozima LENR coupling

Scale: Stellar (106 M? BH - TDE disruption radius)

CP Implementation: ASASSN14liTDEoutflowFUBiCalculator (CondensedPhysics3.py, Session 96)

Testable Prediction: This UQFF result is directly testable with NICER/Chandra (X-ray; testable at 3s by 2027); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.